

Prototyping Therapeutic Gaming: from Semi-Immersive to Fully Immersive VR Experience for Parkinson's Disease

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ABSTRACT

Parkinson's Disease (PD) is a neurodegenerative disorder where physical exercise is crucial for symptom management and quality of life. This paper presents prototypes supporting therapeutic exercises for PD patients: the *Wall Game*, utilizing Kinect, and the *Object Sorting Game*, using Meta Quest. These prototypes target motor coordination, balance, and cognitive engagement by integrating physiotherapy exercises into game scenarios. A user study with 30 participants evaluated usability, cognitive demand, and therapeutic potential. Results showed high usability and low workload scores, confirming these games' feasibility as therapeutic tools. Future work will validate them with PD patients and refine gameplay based on feedback.

Index Terms: Serious Games—User Study—Low-cost Controllers—VR—Parkinson's Disease—Physiotherapy.

1 INTRODUCTION AND BACKGROUND

Parkinson's Disease (PD), the second most common neurodegenerative disorder after Alzheimer's, affects more than 10 million people around the world [6]. Regular physical exercise is vital in managing symptoms to improve motor function, muscle strength, and overall quality of life. However, individuals with PD often face challenges in maintaining exercise routines due to apathy and reduced motivation associated with dopamine deficits [3]. Although effective, traditional rehabilitation methods often fail to sustain long-term patient engagement, emphasizing the need for more engaging and effective solutions.

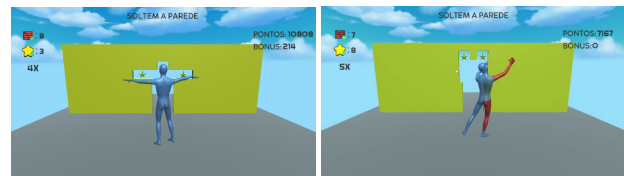
Gamified interventions through serious games (SG) have emerged as promising tools to bridge this gap, using game mechanics to motivate patients and improve adherence to therapeutic exercises [1]. Virtual Reality (VR) further enhances SG by creating immersive gaming environments that promote sustained engagement in treatment regimens [5]. Prior studies have shown the potential of SG to improve the quality of life for PD patients by integrating personalization, challenge, and motivation [2].

This study aims to compare two SG prototypes tailored for PD rehabilitation, each offering distinct levels of immersion and interaction. The *Wall Game*, utilizing Kinect V2, is a semi-immersive game designed to improve balance and coordination, while the *Object Sorting Game*, using the Meta Quest 2 VR headset, provides a fully immersive experience targeting upper limb coordination. Preliminary results suggest that both prototypes are highly usable and effective in addressing therapeutic needs, with differences in physical demand and engagement highlighting complementary strengths.

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(a) Correct T-pose wall

(b) Wrong pose with body feedback

Figure 1: Semi-Immersion VR experience using Kinect

2 PROTOTYPES DESIGN, METHODS AND PARTICIPANTS

Both prototypes were developed using Unity, ensuring compatibility with their respective hardware. Game mechanics were co-designed with physiotherapists to align with therapeutic exercises and ensure accessibility for users with varying levels of motor impairments. The development of the prototypes followed a set of design principles tailored to the needs of people with PD:

- **Trained Functionality:** Incorporating exercises that improve motor and cognitive functions through dual-task challenges, as recommended for therapeutic exercises in PD [3].
- **Feedback and Motivation:** Employing positive reinforcement systems to encourage user engagement without penalizing errors. For instance, maintaining correct posture or completing tasks results in visual and auditory rewards [3].
- **Accessibility:** Designing for diverse physical abilities by incorporating options such as seated modes, simple controls, and intuitive gameplay mechanics. Special attention was given to minimizing cognitive and physical strain for older users [1].
- **Immersion and Relatability:** Creating environments that evoke real-world scenarios or familiar activities, enhancing engagement and emotional connection [5].

The *Wall Game* (Fig. 1) is a semi-immersive serious game designed to improve balance and motor coordination. The game tracks players' body movements, requiring them to mimic specific poses to pass through advancing virtual walls. Inspired by the Japanese television show *Brain Wall*, this game adapts its challenges to physiotherapy goals, encouraging patients to maintain precise postures over time. Misaligned body parts are highlighted in red on the player's avatar, providing immediate feedback to guide corrections. Rewards, such as bonus points for maintaining consecutive successful poses and opportunities to collect interactive items, are designed to encourage sustained engagement.

The *Object Sorting Game* (Fig. 2) provides a fully immersive VR experience. Players interact with virtual objects, catching and sorting them into color-coded bins. The game incorporates increasing difficulty levels, with objects appearing at varying angles and speeds. Cross-body movements are rewarded to encourage coordination and cognitive engagement, making the game suitable for

upper limb rehabilitation. The environment is designed to be visually stimulating while avoiding overwhelming the player.

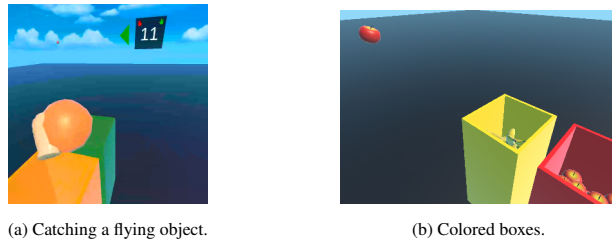


Figure 2: Fully Immersive VR experience using Meta Quest 2

Each session began with an introduction to the study objectives and a brief tutorial on game mechanics. Participants engaged with each prototype for three minutes, starting with either the *Wall Game* or the *Object Sorting Game*, alternating the order to mitigate potential effects of fatigue or learning bias. For the *Wall Game*, participants were asked to complete a predefined sequence of wall challenges. For the *Object Sorting Game*, participants aimed to achieve the highest score by correctly sorting objects under time constraints.

Post-session, participants completed questionnaires that included the System Usability Scale (SUS) and NASA Task Load Index (NASA-TLX) to assess usability and workload, alongside domain-specific Likert-scale questions to evaluate the prototypes' suitability for therapeutic purposes. Statistical analysis was conducted to identify significant differences between the prototypes.

The study involved 30 participants aged 19 to 29 ($M = 22.87$; $SD = 1.99$). 25 identified as male, four as female, and one as another gender. 12 participants had completed secondary school, 14 held a bachelor's degree, and four had obtained a master's degree. The study involved healthy and juvenile participants to ensure that the prototypes' quality and usability reached a stable level before they were introduced to people with PD.

3 RESULTS

Both prototypes achieved high usability scores, with SUS results of 87.7 for the *Wall Game* and 87.2 for the *Object Sorting Game*, placing them in the "excellent" category. NASA-TLX results indicated low overall workload, with mean scores of 23.1 and 18.6, respectively, for the *Wall Game* and the *Object Sorting Game*. These scores fall significantly below the reported mean workload of 45.29 for general tasks in prior studies [4].

To compare the two prototypes, paired statistical tests were conducted. The box plots for the variables with a statistically significant difference ($p < 0.05$) can be seen in Figure 3.

For domain-specific questions, the Wilcoxon signed-rank test was used due to the ordinal nature of the data. A significant difference ($p = 0.02$ between *W-DS1* and *O-DS1*) was found for participants' understanding of the game's goals, with the *Wall Game* scoring higher. However, users reported that they understood better the mapping of their movements in the virtual world in the *Object Sorting Game* ($p = 0.04$ between *W-DS2* and *O-DS2*). This is probably due to the fact that in the fully immersive game only their hands are mapped into the virtual world. There was also a significant difference in the SUS question regarding the learning curve ($p = 0.02$ between *W-SUS10* and *O-SUS10*), with participants reporting a steeper curve for the *Object Sorting Game*.

For NASA-TLX scores, paired t-tests were performed, revealing significant differences in physical demand ($p = 0.0002$, for *W-NP* and *O-NP*) and effort ($p = 0.01$, for *W-NE* and *O-NE*). The *Wall Game* was perceived as more physically demanding, while the *Object Sorting Game* required less effort overall. These findings align

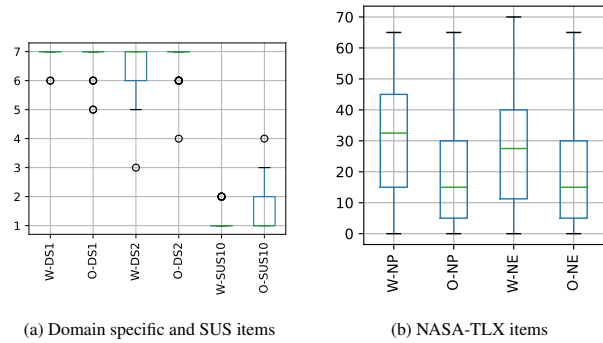


Figure 3: Boxplots of the statistically significant variables ($p < 0.05$). The columns prefixed *W* and *O* are related to the *Wall Game* and *Object Sorting Game*, respectively

with the gameplay mechanics, where the *Wall Game* involves full-body movements, and the *Object Sorting Game* focuses on upper limb coordination.

Qualitative feedback reinforced the quantitative findings. Participants appreciated the *Wall Game* for its intuitive design and focus on posture and balance, which they perceived as directly applicable to physiotherapy. The *Object Sorting Game* was praised for its immersive environment and engaging challenges, despite initial adjustment difficulties with the VR headset. Overall, the statistical and qualitative results suggest that both prototypes are effective in achieving their therapeutic objectives, with complementary strengths that address different aspects of PD rehabilitation.

4 CONCLUSIONS AND FUTURE WORK

These findings provide insights into designing future interventions that balance immersion, usability, and therapeutic outcomes for people with PD. Future research will involve testing the prototypes with people with PD to validate their therapeutic efficacy. Personalization features, such as adaptive difficulty levels and biofeedback integration, will be explored to enhance the effectiveness of these games.

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